

Soil Nutrient Management

Land Sciences

May, 2026

Soil Nutrient Management (A07261)

Short Title: Soil Nutrient Management
Faculty Science and Computing
Department Land Sciences
Cluster Agriculture
Author TWOODCOCK
Credits 5

Level: Advanced

Description of Module / Aims

The module is designed for students who already have a basic knowledge of soils, and will examine in detail the factors affecting soil quality. The movements of nitrogen, phosphorus and potash in and on the soil will be explored, as well as gaseous nitrogen emissions. The importance of soil hydrology will be examined. Soil biology and soil organic matter will be discussed, as well as the potential for carbon dioxide sequestration in grassland. Visits to Teagasc research centres form an integral part of this module.

Programmes

S0IL-0007	BSc (Hons) in Agricultural Science (WD_SAGRI_B)	4 / 8 / M
S0IL-0007	BSc (Hons) in Land Management (in Agriculture) (WD_SBSAG_B)	1 / 2 / M

Indicative Content

- Soil quality and effects on agricultural systems
- The movements of nitrogen, phosphorus and potash in and on the soil and gaseous nitrogen emissions
- Soil hydrology, including soil drainage
- Soil biology, organic matter and carbon dioxide emissions
- National research strategy in the area of soil science
- Nutrient modelling

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Assess the factors affecting soil quality.
2. Compare the movements of nitrogen, phosphorus and potash in and on the soil (including gaseous nitrogen emissions).
3. Determine the factors affecting soil performance.
4. Assess the impact of soil biology and organic matter on carbon dioxide sequestration in grassland.
5. Evaluate the national soil science research strategy.
6. Recommend improvements to soil management strategies which could improve soil performance.

Learning and Teaching Methods

- Lectures.
- Outdoor demonstrations.
- Expert lecturers.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	70%	1,2,4,5
Continuous Assessment	30%	
<i>Project</i>	30%	3,6

Assessment Criteria

<40%: Very limited knowledge and understanding of the subject matter.

40%-49%: Will be able to show a good knowledge of the key learning outcomes including the ability to carry out basic soil tests.

50%-59%: As well as the above the student will be able to integrate the various components of the course.

60%-69%: In addition, the student will demonstrate more detailed knowledge of all the topics, including how to determine what soil tests to use in particular situations.

70%-100%: The learner will be able to demonstrate comprehensive knowledge and understanding of all learning outcomes, including the ability to fully integrate the various components of the course.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Field Trips	12	
Independent Learning	87	

Essential Material

- Bardgett, R. *The Biology of Soil; A community and ecosystem approach*. Oxford: Oxford University Press, 2005.
- Whalen, J.K. *Soil Fertility Improvement and Integrated Nutrient Management - A Global Perspective*. Croatia: InTech, 2012.

Supplementary Material

- Havlin, J.L., J.D. Beaton, S.L. Tisdale and W.L. Nelson. *Soil fertility and fertilizers: an introduction to nutrient management*. 7th ed. Upper Saddle River, N.J.: Pearson Prentice Hall, 2005.